

A group of children and adults are wading in shallow ocean water. In the center, a woman wearing a red cap and a white tank top is looking down at something in her hands. To her right, a man in a dark blue shirt is also looking down. Several children are scattered around them, some looking at the water and others looking towards the camera. The water is shallow and clear, revealing the sandy bottom. In the background, there is a concrete walkway and some greenery.

— A PILOT PROPOSAL FOR JOURNEY SCHOOL · TAMARINDO · 2026

Ocean Literacy at Journey School

A pilot program for school-wide marine science education.

— EFFI BAR · MARINE BIOLOGIST & EDUCATOR

Program overview

A marine biology program design, aligned with **IB PYP and MEP standards** and built around the internationally recognized **Ocean Literacy framework** (NOAA / NMEA).

The program combines hands-on in-school learning and excursions, using our coastline as a living classroom and connecting students directly with marine ecosystems through fieldwork and scientific inquiry.

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Before I show you anything — how do you currently approach marine and environmental education? What's working? What feels missing?

— THE QUESTION I'D LIKE TO START WITH

Why this fits Journey

a.

Standards-aligned, inside the curriculum.

Work together to ensure standards-aligned lessons that fit within the existing curriculum — IB PYP and MEP frameworks, anchored in Ocean Literacy.

CURRICULUM-INTEGRATED

b.

An advantage and big draw for parents.

An Ocean Literacy program in a school is a real differentiator for families — visible, distinctive, and scientifically credible.

ADMISSIONS ADVANTAGE

c.

Career options starting at middle school.

Most students don't know there are careers related to the ocean — boat captain, researcher, and more. The program lets them picture themselves in different future careers.

FUTURE PATHWAYS

Learning objectives

What students will develop across the program.

- i.* Develop ocean literacy — **understanding the ocean's influence on humans** and human impact on the ocean.
- ii.* Build **scientific observation, data collection, and critical thinking skills**.
- iii.* Encourage **environmental stewardship** and responsible decision-making.
- iv.* Connect classroom learning to **real-world marine ecosystems**.

The seven principles

01 Earth has one big ocean with many features.

03 The ocean is a major influence on weather and climate.

05 The ocean supports a great diversity of life and ecosystems.

07 The ocean is largely unexplored.

02 The ocean and life in the ocean shape the features of Earth.

04 The ocean made Earth habitable.

06 The ocean and humans are inextricably interconnected.

Proposed progression by grade

An example scaffolded framework — revised and finalized together with Journey to fit your standards and curriculum.

K-2 <i>"The ocean is alive, and I've met it."</i> Sensory exploration and storytelling. Introduction to marine life and habitats.	GRADES 3-5 <i>"The ocean is a system, and I can describe it."</i> Understanding ecosystems and human impact. Introduction to data collection and scientific vocabulary.	GRADES 6-8 <i>"The ocean has problems, and I can investigate them."</i> Ecosystem dynamics and environmental challenges. Water-quality testing and citizen science.	GRADES 9-12 <i>"The ocean has answers, and I can find them."</i> Research-based learning and field investigations. Marine policy, conservation, and data analysis.
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Instruction methods

Each lesson is 90 minutes. Draws on data sets from NOAA, Monterey Bay Aquarium, Woods Hole, and other leaders in the field.



Hands-on activities

Build a rocky shore model. Scavenger hunts. Specimen observation. Plankton towing.



Field excursions

Tide pools at Avellanas / Negra. Sea turtle nesting. Mangrove and estuary exploration. Marine agriculture in the Gulf of Nicoya.



Data collection and logs

Water clarity. Beach debris counts. Plankton sampling. Microplastics analysis. Ecological survey and citizen science.



Worksheets & structured discussion

Worksheets, structured experiments, and class discussions throughout.

Two ways in

OPTION 1 · RECOMMENDED

A 1-year program, beginning in December.

Based on 9–10 months of school.

- Monthly visits with a marine biologist
- 1 field excursion per semester (3 total)
- 9 × 90-minute hands-on, PBL lessons
- Worksheets, field guides, and classroom materials included
- Differentiation strategies to support diverse learners
- 2–3 professional development sessions (Train-the-Trainer)

OPTION 2 · LIGHTER TOUCH

One project over a few months.

- One ocean literacy project per grade level
- 1 field excursion
- Number of lessons TBD with the program
- Project-based assessments
- Worksheets, field guides, and classroom materials included
- Differentiation strategies to support diverse learners
- 2 teacher training sessions

About Effi



Effi (Efrat Bar) — marine biologist and environmental educator, with over a decade of experience.

MARINE SCIENCE · CONSERVATION · EDUCATION

B.Sc. in Marine Sciences (with honors); graduate studies in environmental studies, focusing on marine protected area management.

- Marine research and conservation work with EcoOcean — ecological surveys, biodiversity documentation, marine reserve planning.
- Educational program development and instruction, in classroom and field settings, including aboard research vessels.
- Public education and outreach with a sea turtle rescue and rehabilitation center.

“Combines scientific expertise with hands-on teaching experience — engaging, curriculum-aligned marine science education tailored to school environments.”

Let's discuss

01 Which **age band(s)** would you prioritize for the pilot?

02 Which **option** fits Journey best — full year, or single project?

03 What **shape** works in your calendar — weekly · special days · camps · transversal?

04 Would you like a **tailored proposal** this week?



I want this built for Journey specifically — not a template.



Thank you.

Questions?

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